

NASA HTTP Log Analytics Tool

Phase 1 Project Report

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Abstract

This report details the Phase 1 implementation of a multi-backend log analytics system designed to process and analyze massive web access logs. Utilizing the standard 1995 NASA HTTP access log dataset, we developed a robust Java-based parsing engine and established two primary analytical pipelines: **MongoDB** (leveraging the Aggregation Framework) and **Apache Hadoop MapReduce** (running in local standalone mode). Furthermore, a persistent reporting layer was constructed using **MySQL** to store query results in a normalized, schema-enforced structure. This document outlines the system architecture, specific query implementations, and comparative benchmarking results.

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1 Introduction and Dataset

The objective of this project is to build an interactive, pipeline-agnostic Command Line Interface (CLI) tool for big data analytics. The dataset utilized is the NASA Kennedy Space Center HTTP server access log from July and August 1995.

1.1 Dataset Characteristics

- **Total Records Processed:** 3,461,613 lines.
- **Format:** Common Log Format with variations (e.g., missing HTTP versions, malformed bytes).
- **Data Cleaning:** A robust Java Regular Expression (`Pattern.compile`) parser was implemented to handle anomalies. Out of 3.46 million records, only exactly 1 record was identified as unrecoverably malformed, demonstrating the parser's high resilience. Data is loaded in configurable batches (default 50,000) to optimize memory consumption.

2 System Architecture

The application is structured using a decoupled "Plug-and-Play" architecture, enabling runtime switching between database backends.

1. **Core CLI & Parser:** Acts as the orchestration layer, reading `.gz` log files, parsing them into `LogRecord` POJOs, and managing batch dispatching.
2. **Database Abstraction (`Database.java`):** A unified interface defining methods for data loading (`loadData`) and analytical queries (`executeDailyTrafficSummary`, etc.).
3. **Persistence Store (`MySQLStore.java`):** A JDBC-connected service that archives query outputs and metadata into relational tables.

3 Pipeline Implementation Details

The project mandates the execution of three distinct analytical queries:

1. **Query 1:** Daily Traffic Summary (Total Requests and Total Bytes per day per status code).
2. **Query 2:** Top 20 Requested Resources (Ranked by Request Count, including Total Bytes and Distinct Host counts).
3. **Query 3:** Hourly Error Analysis (Tracking 4xx/5xx HTTP errors, total requests, and error rates per hour).

3.1 Pipeline A: MongoDB Aggregation Framework

The MongoDB implementation ingests the parsed `LogRecord` objects directly into a `logs` collection. Queries are constructed using the BSON Aggregation Pipeline.

- **Query 1 (Daily Traffic):** Utilizes a `$group` stage matching on a compound `_id` of `{ logDate, statusCode }`. It uses `$sum` accumulators to count records and sum `bytesTransferred`. A subsequent `$sort` stage orders the output chronologically.
- **Query 2 (Top 20 Resources):** Employs `$group` on `resourcePath`. To calculate the unique visitors per resource, the `$addToSet` operator is used to aggregate unique IP/hostnames into an array, whose size is later projected. A `$sort` on request count descending and a `$limit` of 20 finalizes the pipeline.
- **Query 3 (Hourly Errors):** Uses a `$project` stage with a `$cond` (conditional) operator to flag records where `statusCode ≥ 400`. It then groups by `logDate` and `logHour` to compute error rates.

3.2 Pipeline B: Apache Hadoop MapReduce (Local Mode)

The MapReduce pipeline operates independently of HDFS, utilizing Hadoop’s standalone mode to process a temporary TSV file generated by the loader.

- **Data Serialization:** To avoid complex custom `Writable` classes, composite keys and values are serialized as tab-separated `Text` objects.
- **Query 1 Implementation:** The Mapper emits (`logDate + statusCode`) as the key and (`1 + bytes`) as the value. A Combiner (reusing the Reducer logic) is applied to perform local aggregation, significantly reducing the data volume processed during the Shuffle phase.
- **Query 2 Implementation:** The Mapper extracts `resourcePath` as the key and (`bytes + host`) as the value. The Reducer uses a Java `HashSet` to track unique hosts per resource. The final sorting to find the "Top 20" is handled in-memory within the Java driver after the Reducer outputs the raw aggregated list.
- **Query 3 Implementation:** The Mapper emits (`logDate + logHour`) and a binary flag indicating if the status code represents an error. The Reducer tallies total vs. error requests to calculate the exact error rate.

4 Relational Schema & Persistence Strategy

To satisfy academic requirements and ensure data integrity, a **One-Table-Per-Query** schema was adopted in MySQL rather than relying on unstructured JSON blobs.

- `run_metadata`: Archives the execution context. Columns include `pipeline_name`, `run_id`, `execution_time`, `batch_size`, `total_records`, and `malformed_count`.
- `query1_results`: Columns for `log_date`, `status_code`, `request_count`, and `total_bytes`.

- `query2_results`: Columns for `resource_path`, `request_count`, `total_bytes`, and `distinct_host_count`.
- `query3_results`: Columns for `log_date`, `log_hour`, `error_request_count`, `total_request_count`, `error_rate`, and `distinct_error_hosts`.

A custom ASCII-table exporter was developed to automatically extract these relational tables into a formatted text file (`query_results_output.txt`) upon application exit.

5 Benchmarking and Performance Results

Both pipelines were benchmarked on the full dataset of 3.46 million records. MongoDB processed data natively in-memory, while MapReduce simulated distributed file I/O locally.

Query Type	MongoDB (s)	MapReduce (s)	Speedup Factor
Q1: Daily Traffic	0.997	5.694	~5.7x
Q2: Top Resources	2.379	5.272	~2.2x
Q3: Hourly Errors	1.054	5.229	~4.9x

Table 1: Query Execution Time Comparison (Batch Size: 50,000)

Analysis: MongoDB drastically outperforms local MapReduce across all queries. This is expected, as MapReduce suffers from significant JVM startup overhead, disk I/O serialization, and framework bootstrapping even for local jobs, whereas MongoDB executes highly optimized native C++ aggregation pipelines over indexed or cached memory collections. Query 2 takes the longest in both systems due to the high cardinality of the distinct host calculation (`$addToSet` in Mongo and `HashSet` in MapReduce).

6 Screenshots Appendix

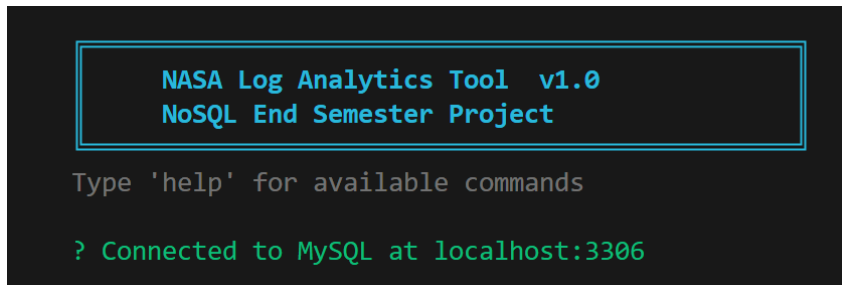


Figure 1: Connection Initialization

```

nasa> use mongodb
? Switched to MongoDB pipeline

nasa> load
? Loading 2 file(s) with batch size 10000...
Batch 50/347 (500,000 records so far)
Batch 100/347 (1,000,000 records so far)
Batch 150/347 (1,500,000 records so far)
Batch 200/347 (2,000,000 records so far)
Batch 250/347 (2,500,000 records so far)
Batch 300/347 (3,000,000 records so far)
Batch 347/347 (3,461,613 records so far)
? Loaded 3,461,613 records in 347 batches (124.8s)
? Avg batch size: 9975.83 | Malformed records: 1 | Run ID: mongodb-20260428-125320

```

Figure 2: Mongo connection

```

nasa> run all
? Running Daily Traffic Summary on MongoDB...

```

Daily Traffic Summary (5.564s on MongoDB)			
log_date	status_code	request_count	total_bytes
1995-07-01	200	58,033	1,617,409,574
1995-07-01	302	2,568	218,465
1995-07-01	304	3,797	0
1995-07-01	404	316	0
1995-07-02	200	54,492	1,531,584,293
1995-07-02	302	2,370	195,795
1995-07-02	304	3,107	0
1995-07-02	403	5	0
1995-07-02	404	291	0
1995-07-03	200	81,351	2,085,066,566
1995-07-03	302	3,122	261,797
1995-07-03	304	4,582	0
1995-07-03	403	2	0
1995-07-03	404	474	0
1995-07-03	500	53	0
1995-07-04	200	63,069	1,744,266,570
1995-07-04	302	2,513	206,813
1995-07-04	304	4,507	0
1995-07-04	403	4	0
1995-07-04	404	359	0
1995-07-05	200	85,080	2,050,959,436
1995-07-05	302	3,349	282,418
1995-07-05	304	5,646	0
1995-07-05	403	2	0
1995-07-05	404	497	0
1995-07-05	501	1	0
1995-07-06	200	91,643	2,218,747,557
1995-07-06	302	3,197	264,274
1995-07-06	304	5,475	0
1995-07-06	403	5	0

```

... and 265 more rows

Total: 295 rows
? Results stored in MySQL (run_id: mongodb-20260428-125320)

```

Figure 3: Mongoddb Query1

? Running Top 20 Requested Resources on MongoDB...

Top 20 Requested Resources (18.554s on MongoDB)			
resource_path	request_count	total_bytes	distinct_host_count
/images/NASA-logosmall.gif	208,798	131,778,402	81,518
/images/KSC-logosmall.gif	164,976	169,364,272	85,161
/images/MOSAIC-logosmall.gif	127,916	40,594,653	54,904
/images/USA-logosmall.gif	127,082	25,993,422	54,498
/images/WORLD-logosmall.gif	125,933	73,796,052	54,054
/images/ksclogo-medium.gif	121,580	629,750,296	49,491
/ksc.html	83,918	563,742,661	29,191
/images/launch-logo.gif	76,009	114,690,489	42,558
/history/apollo/images/apollo-logo1.gif	68,898	71,784,081	38,406
/shuttle/countdown/	64,740	255,164,345	36,898
/	63,268	400,138,583	31,819
/images/ksclogosmall.gif	61,395	202,018,760	38,842
/shuttle/missions/missions.html	47,318	379,528,351	27,564
/images/launchmedium.gif	40,689	445,755,771	24,844
/htbin/cdt_main.pl	39,873	143,972,596	12,661
/shuttle/missions/sts-69/mission-sts-69.html	31,574	317,318,518	18,175
/shuttle/countdown/liftoff.html	29,865	112,077,906	14,564
/icons/menu.xbm	29,190	13,909,111	17,819
/shuttle/missions/sts-69/sts-69-patch-small.gif	29,119	212,174,151	18,070
/icons/blank.xbm	28,852	13,214,149	17,846

Total: 20 rows
 ? Results stored in MySQL (run_id: mongodb-20260428-125320)

Figure 4: Mongoddb Query2

? Running Hourly Error Analysis on MongoDB...

Hourly Error Analysis (7.424s on MongoDB)					
log_date	log_hour	error_request_count	total_request_count	error_rate	distinct_error_hosts
1995-07-01	0	24	3,565	0.006732	12
1995-07-01	1	10	3,004	0.003329	7
1995-07-01	2	12	2,268	0.005291	9
1995-07-01	3	16	1,734	0.009227	5
1995-07-01	4	10	1,482	0.006748	8
1995-07-01	5	9	1,343	0.006701	5
1995-07-01	6	4	1,528	0.002618	3
1995-07-01	7	2	1,557	0.001285	1
1995-07-01	8	6	1,927	0.003114	4
1995-07-01	9	3	2,096	0.001431	2
1995-07-01	10	13	3,046	0.004268	7
1995-07-01	11	23	3,428	0.006709	13
1995-07-01	12	10	2,769	0.003611	7
1995-07-01	13	13	3,086	0.004213	10
1995-07-01	14	19	3,133	0.006064	14
1995-07-01	15	17	3,272	0.005196	12
1995-07-01	16	14	3,191	0.004387	9
1995-07-01	17	15	3,671	0.004086	9
1995-07-01	18	15	3,192	0.004699	11
1995-07-01	19	16	2,831	0.005652	11
1995-07-01	20	10	2,958	0.003381	8
1995-07-01	21	24	3,502	0.006853	12
1995-07-01	22	18	3,278	0.005491	12
1995-07-01	23	13	2,853	0.004557	7
1995-07-02	0	20	2,858	0.006998	11
1995-07-02	1	15	2,570	0.005837	9
1995-07-02	2	8	2,056	0.003891	4
1995-07-02	3	7	1,505	0.004651	5
1995-07-02	4	2	1,286	0.001555	2
1995-07-02	5	3	1,291	0.002324	3

... and 1339 more rows
 Total: 1369 rows
 ? Results stored in MySQL (run_id: mongodb-20260428-125320)

Figure 5: Mongoddb Query3

```

nasa> use mr
? Switched to MapReduce pipeline

nasa> load
? Loading 2 file(s) with batch size 10000...
Batch 50/347 (500,000 records so far)
Batch 100/347 (1,000,000 records so far)
Batch 150/347 (1,500,000 records so far)
Batch 200/347 (2,000,000 records so far)
Batch 250/347 (2,500,000 records so far)
Batch 300/347 (3,000,000 records so far)
Batch 347/347 (3,461,613 records so far)
? Loaded 3,461,613 records in 347 batches (16.9s)
? Avg batch size: 9975.83 | Malformed records: 1 | Run ID: mapreduce-20260428-125633

```

Figure 6: MapReduce connection

```

nasa> run all
? Running Daily Traffic Summary on MapReduce...

```

log_date	status_code	request_count	total_bytes
1995-07-01	200	58,033	1,617,409,574
1995-07-01	302	2,568	218,465
1995-07-01	304	3,797	0
1995-07-01	404	316	0
1995-07-02	200	54,492	1,531,584,293
1995-07-02	302	2,370	195,795
1995-07-02	304	3,107	0
1995-07-02	403	5	0
1995-07-02	404	291	0
1995-07-03	200	81,351	2,085,066,566
1995-07-03	302	3,122	261,797
1995-07-03	304	4,582	0
1995-07-03	403	2	0
1995-07-03	404	474	0
1995-07-03	500	53	0
1995-07-04	200	63,069	1,744,266,570
1995-07-04	302	2,513	206,813
1995-07-04	304	4,507	0
1995-07-04	403	4	0
1995-07-04	404	359	0
1995-07-05	200	85,080	2,050,959,436
1995-07-05	302	3,349	282,418
1995-07-05	304	5,646	0
1995-07-05	403	2	0
1995-07-05	404	497	0
1995-07-05	501	1	0
1995-07-06	200	91,643	2,218,747,557
1995-07-06	302	3,197	264,274
1995-07-06	304	5,475	0
1995-07-06	403	5	0

```

... and 265 more rows
Total: 295 rows
? Results stored in MySQL (run_id: mapreduce-20260428-125633)

```

Figure 7: MapReduce Query1

? Running Top 20 Requested Resources on MapReduce...

Top 20 Requested Resources (15.424s on MapReduce)			
resource_path	request_count	total_bytes	distinct_host_count
/images/NASA-logosmall.gif	208,798	131,778,402	81,518
/images/KSC-logosmall.gif	164,976	169,364,272	85,161
/images/MOSAIC-logosmall.gif	127,916	40,594,653	54,904
/images/USA-logosmall.gif	127,082	25,993,422	54,498
/images/WORLD-logosmall.gif	125,933	73,796,052	54,054
/images/ksclogo-medium.gif	121,580	629,750,296	49,491
/ksc.html	83,918	563,742,661	29,191
/images/launch-logo.gif	76,009	114,690,489	42,558
/history/apollo/images/apollo-logo1.gif	68,898	71,784,081	38,406
/shuttle/countdown/	64,740	255,164,345	36,898
/	63,268	400,138,583	31,819
/images/ksclogosmall.gif	61,395	202,018,760	38,842
/shuttle/missions/missions.html	47,318	379,528,351	27,564
/images/launchmedium.gif	40,689	445,755,771	24,844
/htbin/cdt_main.pl	39,873	143,972,596	12,661
/shuttle/missions/sts-69/mission-sts-69.html	31,574	317,318,518	18,175
/shuttle/countdown/liftoff.html	29,865	112,077,906	14,564
/icons/menu.xbm	29,190	13,909,111	17,819
/shuttle/missions/sts-69/sts-69-patch-small.gif	29,119	212,174,151	18,070
/icons/blank.xbm	28,852	13,214,149	17,846

Total: 20 rows
 ? Results stored in MySQL (run_id: mapreduce-20260428-125633)

Figure 8: MapReduce Query2

? Running Hourly Error Analysis on MapReduce...

Hourly Error Analysis (15.316s on MapReduce)					
log_date	log_hour	error_request_count	total_request_count	error_rate	distinct_error_hosts
1995-07-01	0	24	3,565	0.006732	12
1995-07-01	1	10	3,004	0.003329	7
1995-07-01	2	12	2,268	0.005291	9
1995-07-01	3	16	1,734	0.009227	5
1995-07-01	4	10	1,482	0.006748	8
1995-07-01	5	9	1,343	0.006701	5
1995-07-01	6	4	1,528	0.002618	3
1995-07-01	7	2	1,557	0.001285	1
1995-07-01	8	6	1,927	0.003114	4
1995-07-01	9	3	2,096	0.001431	2
1995-07-01	10	13	3,046	0.004268	7
1995-07-01	11	23	3,428	0.006709	13
1995-07-01	12	10	2,769	0.003611	7
1995-07-01	13	13	3,086	0.004213	10
1995-07-01	14	19	3,133	0.006064	14
1995-07-01	15	17	3,272	0.005196	12
1995-07-01	16	14	3,191	0.004387	9
1995-07-01	17	15	3,671	0.004086	9
1995-07-01	18	15	3,192	0.004699	11
1995-07-01	19	16	2,831	0.005652	11
1995-07-01	20	10	2,958	0.003381	8
1995-07-01	21	24	3,502	0.006853	12
1995-07-01	22	18	3,278	0.005491	12
1995-07-01	23	13	2,853	0.004557	7
1995-07-02	0	20	2,858	0.006998	11
1995-07-02	1	15	2,570	0.005837	9
1995-07-02	2	8	2,056	0.003891	4
1995-07-02	3	7	1,505	0.004651	5
1995-07-02	4	2	1,286	0.001555	2
1995-07-02	5	3	1,291	0.002324	3

... and 1339 more rows
 Total: 1369 rows
 ? Results stored in MySQL (run_id: mapreduce-20260428-125633)

Figure 9: MapReduce Query3

nasa> report

Stored Query Results

Pipeline: MapReduce Query: query3
Run ID: mapreduce-20260428-125633 Exec Time: 15.316s
Batch Size: 10000 Batches: 347
Total Recs: 3461613 Avg Batch: 9975.83
Malformed: 1 Time: 2026-04-28 12:57:38.0

Pipeline: MapReduce Query: query2
Run ID: mapreduce-20260428-125633 Exec Time: 15.424s
Batch Size: 10000 Batches: 347
Total Recs: 3461613 Avg Batch: 9975.83
Malformed: 1 Time: 2026-04-28 12:57:23.0

Pipeline: MapReduce Query: query1
Run ID: mapreduce-20260428-125633 Exec Time: 11.244s
Batch Size: 10000 Batches: 347
Total Recs: 3461613 Avg Batch: 9975.83
Malformed: 1 Time: 2026-04-28 12:57:07.0

Pipeline: MongoDB Query: query3
Run ID: mongodb-20260428-125320 Exec Time: 7.424s
Batch Size: 10000 Batches: 347
Total Recs: 3461613 Avg Batch: 9975.83
Malformed: 1 Time: 2026-04-28 12:56:09.0

Pipeline: MongoDB Query: query2
Run ID: mongodb-20260428-125320 Exec Time: 18.554s
Batch Size: 10000 Batches: 347
Total Recs: 3461613 Avg Batch: 9975.83
Malformed: 1 Time: 2026-04-28 12:56:01.0

Pipeline: MongoDB Query: query1
Run ID: mongodb-20260428-125320 Exec Time: 5.564s
Batch Size: 10000 Batches: 347
Total Recs: 3461613 Avg Batch: 9975.83
Malformed: 1 Time: 2026-04-28 12:55:43.0

Figure 10: Final Report